

SIMATS ENGINEERING

ASSESSMENT TOOL II — DATA ANALYSIS PROJECT

Course: Data Handling and Visualization	Code: DSA0603	CO: CO3
Duration: 180 min	Total Marks: 100	Reg No : 192524029 Name: KAILASH M

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Data Analysis Project

Weightage: 20%

Code of Conduct:

I KAILASH M (192524029) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: KAILASH M

Criteria	Understanding of Concepts 25%	Analytical Design 25%	Use of Tools 20%	Analysis of Results 20%	Documentation 10%	Total
Q1						
Q2						
Q3						
Q4						
Q5						

DSA0603 — DATA HANDLING AND VISUALIZATION

CO3_AT2_Assessment Tool: DATA ANALYSIS PROJECT

Topics Covered: ECDF and Q-Q Plots; Multiple Distributions; Vertical-Axis Distributions; Bean Plots; Proportions; Pie/Side-by-Side/Stacked Bars; Stacked Densities; Treemaps; Sunburst Charts; Parallel Sets; Sankey Diagrams; Nested Proportions; Avoiding Misleading Proportions.

General Instruction: For each project task, use an appropriate visualization in R, justify the design choice, interpret the result in context, and identify potential visualization bias. R/ggplot2 (or an equivalent tool such as treemapify, ggalluvial, or networkD3) may be used. Submit R code, output plots, and a short written interpretation for each question.

Q1 — ECDF and Q-Q Plots

```
# Q1: Comparing Distributions with ECDF and Q-Q Plots

# Data
partner_a <- c(18,19,20,20,21,21,22,22,23,23,
               24,24,25,26,27,28,30,33,38,46)

partner_b <- c(20,21,22,22,23,23,23,24,24,24,
               24,25,25,25,26,26,27,27,28,29)

data <- data.frame(
  Time = c(partner_a, partner_b),
  Partner = rep(c("Partner A", "Partner B"), each = 20)
)

# 1. ECDF Plot
library(ggplot2)

ggplot(data, aes(x = Time, color = Partner)) +
  stat_ecdf(linewidth = 1.2) +
  labs(
    title = "ECDF of Delivery Times",
    x = "Delivery Time (hours)",
    y = "Cumulative Proportion"
  ) +
  theme_minimal()

# 2. Q-Q Plot for Partner A
qqnorm(partner_a,
        main = "Normal Q-Q Plot - Partner A",
        pch = 19)
qqline(partner_a, col = "red", lwd = 2)

# 3. Q-Q Plot for Partner B
qqnorm(partner_b,
        main = "Normal Q-Q Plot - Partner B",
        pch = 19)
qqline(partner_b, col = "red", lwd = 2)
```

```

# 4. Find maximum ECDF divergence

ecdf_a <- ecdf(partner_a)
ecdf_b <- ecdf(partner_b)

thresholds <- sort(unique(c(partner_a, partner_b)))

result <- data.frame(
  Time = thresholds,
  Partner_A = ecdf_a(thresholds),
  Partner_B = ecdf_b(thresholds)
)

result$Difference <- abs(
  result$Partner_A - result$Partner_B
)

print(result)

max_difference <- result[
  which.max(result$Difference), ]

print(max_difference)

# 5. Mean and Standard Deviation

cat("Partner A Mean =", mean(partner_a), "\n")
cat("Partner A SD =", sd(partner_a), "\n")

cat("Partner B Mean =", mean(partner_b), "\n")
cat("Partner B SD =", sd(partner_b), "\n")

# 6. 90th Percentile

cat("Partner A 90th Percentile =",
  quantile(partner_a, 0.90), "\n")

cat("Partner B 90th Percentile =",
  quantile(partner_b, 0.90), "\n")

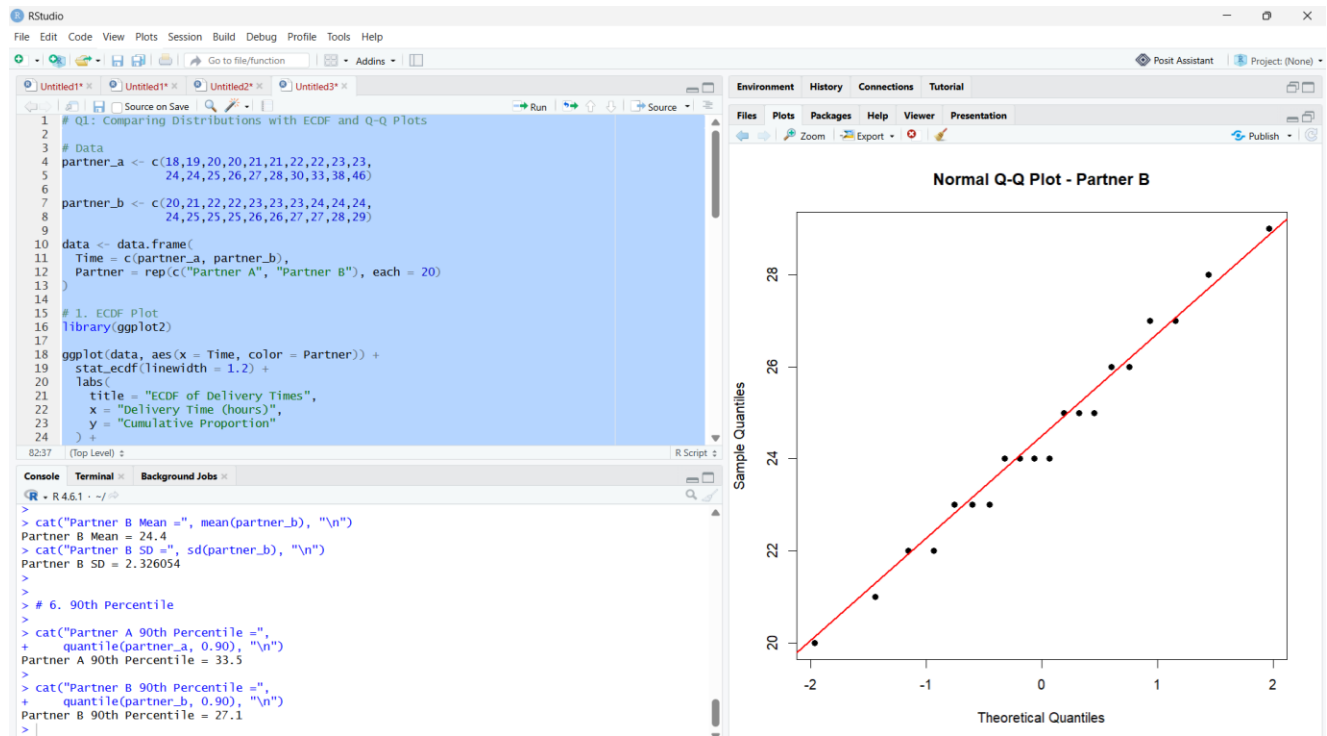
```

Q1 Interpretation

ECDF: Partner B is generally faster and more consistent. Partner A has a longer upper tail, especially because of the 38-hour and 46-hour deliveries.

Q-Q plots: Partner A shows stronger deviation from the reference line at the upper end, indicating positive skew and possible outliers. Partner B is closer to a normal pattern.

SLA recommendation: Empirical percentile-based SLAs are preferable to mean $\pm$ standard deviation because Partner A contains extreme delivery times and the distributions are not perfectly normal.



Q2 — Bean Plots / Distribution Shape

```
# Q2: Distribution Shape Across Many Groups

library(ggplot2)
library(dplyr)

department <- c(
  rep("Cardiology", 10),
  rep("Orthopedics", 10),
  rep("Pediatrics", 10),
  rep("Dermatology", 10),
  rep("General Medicine", 10)
)

waiting <- c(
  10,12,13,14,15,15,16,17,18,20,
  8,9,25,26,27,9,10,28,29,8,
  5,6,6,7,7,8,8,9,9,10,
  15,16,17,18,19,20,21,22,23,24,
  10,40,12,38,11,41,13,39,12,40
)

data <- data.frame(
  Department = department,
  WaitingTime = waiting
)

# 1. Bean Plot Equivalent
ggplot(data, aes(x = Department, y = WaitingTime)) +
  geom_violin(trim = FALSE, alpha = 0.5) +
  geom_jitter(width = 0.12, size = 2) +
  labs(
    title = "Patient Waiting Time Distribution",
    x = "Department",
    y = "Waiting Time (minutes)"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(
      angle = 30,
      hjust = 1
    )
  )

# 2. Boxplot

ggplot(data, aes(x = Department, y = WaitingTime)) +
  geom_boxplot() +
  labs(
    title = "Boxplot of Patient Waiting Times",
    x = "Department",
    y = "Waiting Time (minutes)"
  )
```

```

) +
theme_minimal() +
theme(
  axis.text.x = element_text(
    angle = 30,
    hjust = 1
  )
)

# 3. Summary statistics

summary_data <- data %>%
  group_by(Department) %>%
  summarise(
    Mean = mean(WaitingTime),
    Median = median(WaitingTime),
    SD = sd(WaitingTime),
    Minimum = min(WaitingTime),
    Maximum = max(WaitingTime),
    Range = max(WaitingTime) - min(WaitingTime)
  )

print(summary_data)

# 4. Variability ranking

cat("VARIABILITY RANKING:\n")

print(
  summary_data %>%
    arrange(desc(SD))
)

# 5. Median ranking

cat("MEDIAN RANKING:\n")

print(
  summary_data %>%
    arrange(desc(Median))
)

```

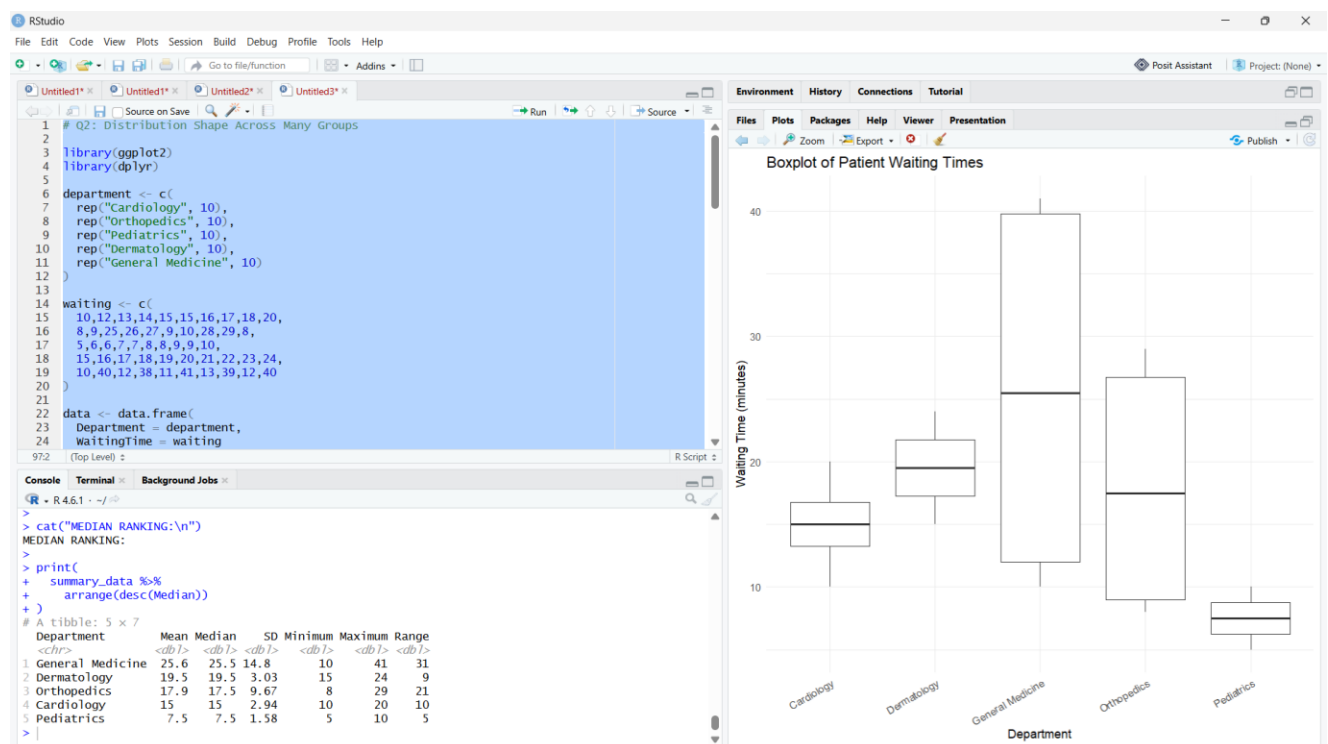
Q2 Interpretation

General Medicine and **Orthopedics** show bimodal patterns.

General Medicine has one group around **10–13 minutes** and another around **38–41 minutes**.
Orthopedics has one group around **8–10 minutes** and another around **25–29 minutes**.

The violin/bean-style plot reveals the distribution shape and multiple modes, while a boxplot mainly shows the median, quartiles and possible outliers.

A useful staffing decision would be to provide **additional staff during periods associated with the long waiting-time group**, particularly in General Medicine.



Q3 — Proportion Charts

```
# Q3: Choosing an Honest Proportion Chart
```

```
library(ggplot2)
```

```
library(tidyr)
```

```
data <- data.frame(  
  Store = c(  
    "Hypermarket",  
    "Supermarket",  
    "Convenience"  
  ),  
  Cash = c(18, 22, 35),  
  Card = c(34, 30, 25),  
  UPI_Wallet = c(48, 48, 40)  
)
```

```
long_data <- data %>%  
  pivot_longer(  
    cols = c(Cash, Card, UPI_Wallet),  
    names_to = "Payment",  
    values_to = "Percentage"  
  )
```

```
# 1. Grouped Bar Chart
```

```
ggplot(  
  long_data,  
  aes(  
    x = Store,  
    y = Percentage,  
    fill = Payment  
  )  
) +  
  geom_col(position = "dodge") +  
  labs(  
    title = "Payment Method by Store Format",  
    x = "Store Format",  
    y = "Percentage (%)"  
  ) +  
  theme_minimal()
```

```
# 2. 100% Stacked Bar Chart
```

```
ggplot(  
  long_data,  
  aes(  
    x = Store,  
    y = Percentage,  
    fill = Payment  
  )  
) +
```



```

geom_col(position = "fill") +
scale_y_continuous(
  labels = function(x) paste0(x * 100, "%")
) +
labs(
  title = "100% Stacked Payment Distribution",
  x = "Store Format",
  y = "Percentage"
) +
theme_minimal()

```

3. Pie Charts

```

ggplot(
  long_data,
  aes(
    x = "",
    y = Percentage,
    fill = Payment
  )
) +
geom_col(width = 1) +
coord_polar(theta = "y") +
facet_wrap(~Store) +
labs(
  title = "Payment Preferences by Store Format"
) +
theme_void()

```

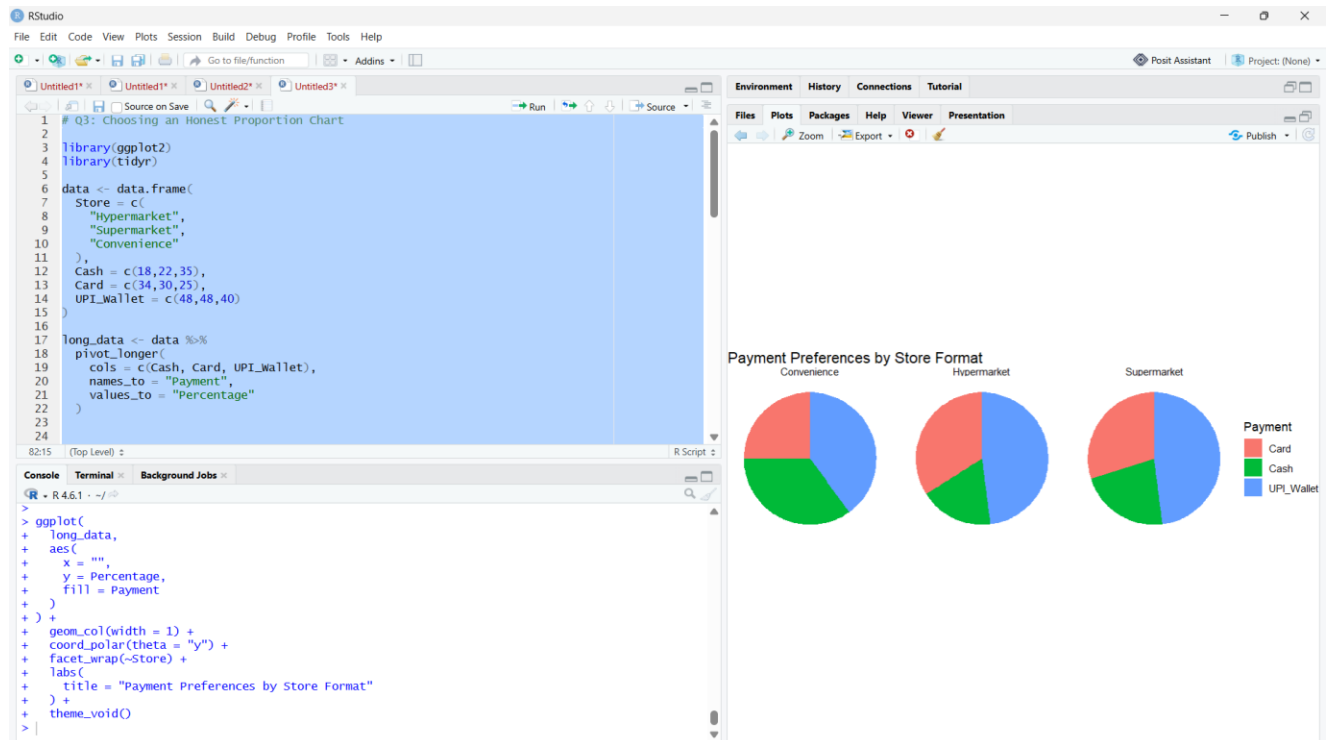
Q3 Interpretation

Hypermarket and Supermarket have the highest UPI/Wallet share at **48%**. Convenience has **40%**.

The grouped bar chart is easiest for accurate comparison because all bars share a common baseline.

Pie charts are harder to compare because people are less accurate at comparing **angles and arc lengths** than comparing bar lengths.

A misleading design would be to **truncate the y-axis**, for example starting it at 30% instead of 0%. This would make the Convenience cash share of 35% appear much larger than it really is.



Q4 — Treemap and Sunburst

```
# Q4: Nested Proportions in a Household Budget
```

```
housing <- sum(
  data$Amount[
    data$Category == "Housing"
  ]
)
```

```
rent <- 18000
```

```
housing_percentage <- housing / total * 100
```

```
rent_percentage <- rent / total * 100
```

```
cat(
  "Total Budget = Rs.",
  total,
  "\n"
)
```

```
cat(
  "Housing = Rs.",
  housing,
  "\n"
)
```

```
cat(
  "Housing Percentage =",
  housing_percentage,
  "%\n"
)
```

```
cat(
  "Rent/EMI Percentage =",
  rent_percentage,
  "%\n"
)
```

```
# 3. Sunburst-style chart
```

```
category_total <- data %>%
  group_by(Category) %>%
  summarise(Amount = sum(Amount)) %>%
  mutate(
    ymax = cumsum(Amount),
    ymin = lag(ymax, default = 0)
  )
```

```
subcategory <- data %>%
  group_by(Category) %>%
  mutate(
    ymax = cumsum(Amount),
    ymin = lag(ymax, default = 0)
  )
```

```

) %>%
ungroup()

ggplot() +
  geom_rect(
    data = category_total,
    aes(
      xmin = 0.5,
      xmax = 1.5,
      ymin = ymin,
      ymax = ymax,
      fill = Category
    )
  ) +
  geom_rect(
    data = subcategory,
    aes(
      xmin = 1.5,
      xmax = 2.5,
      ymin = ymin,
      ymax = ymax,
      fill = Category
    )
  ) +
  coord_polar(theta = "y") +
  xlim(0, 2.5) +
  labs(
    title = "Household Budget Sunburst"
  ) +
  theme_void()

```

Q4 Interpretation

Total budget = **₹60,000**

Housing:

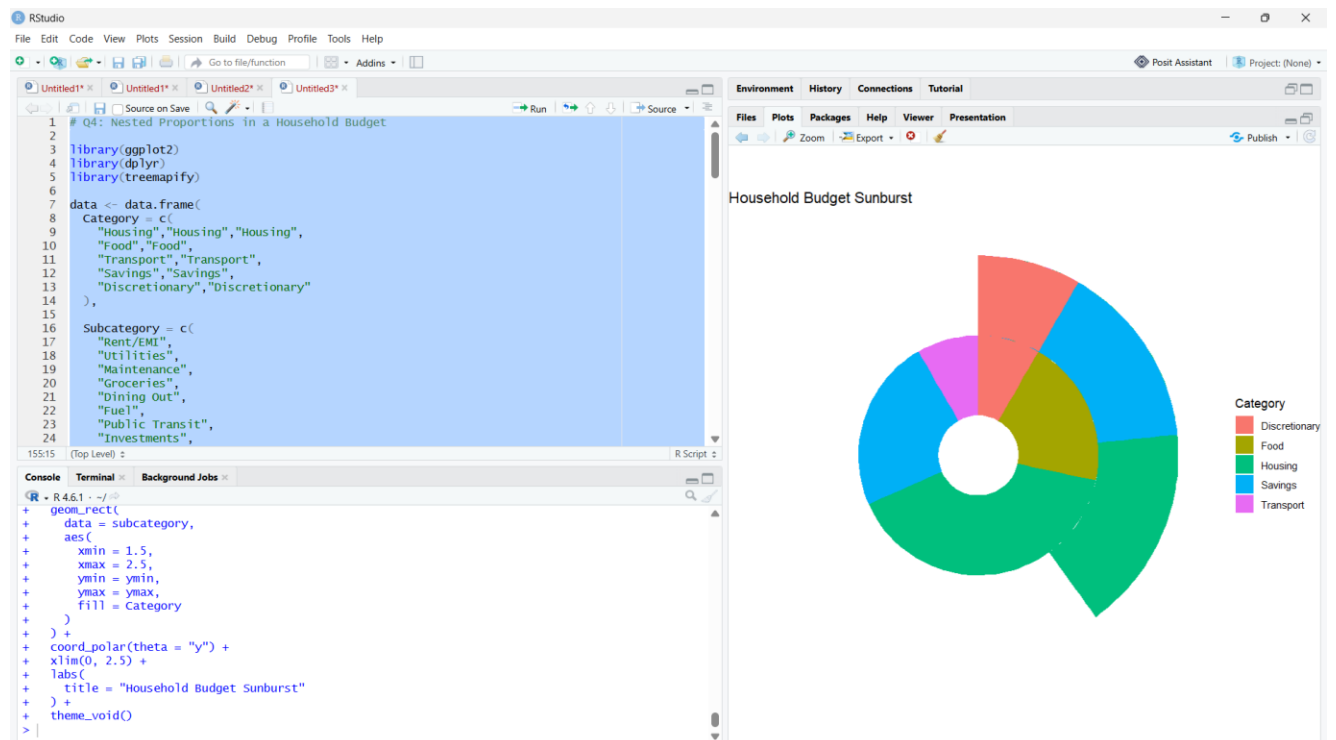
₹24,000 / ₹60,000 × 100 = 40%

Largest Housing subcategory:

Rent/EMI = ₹18,000 / ₹60,000 × 100 = 30%

The treemap makes area comparison easy and shows the hierarchy. The sunburst is useful for showing the relationship between categories and subcategories.

However, comparing subcategories from different categories is generally easier in a treemap than in a sunburst because rectangular areas are easier to compare than curved segments.



Q5 — Sankey and Parallel Sets

```
# Q5: Flow Analysis
)

links <- rbind(link1, link2)

# Sankey

sankeyNetwork(
  Links = links,
  Nodes = nodes,
  Source = "source",
  Target = "target",
  Value = "value",
  NodeID = "name",
  fontSize = 14,
  nodeWidth = 30,
  sinksRight = TRUE
)

# -----
# 2. Parallel Sets / Alluvial Plot
# -----

ggplot(
  data,
  aes(
    axis1 = Course,
    axis2 = Device,
    axis3 = Status,
    y = Students
  )
) +
  geom_alluvium(
    aes(fill = Course),
    width = 1/12,
    alpha = 0.7
  ) +
  geom_stratum(width = 1/8) +
  geom_text(
    stat = "stratum",
    aes(label = after_stat(stratum))
  ) +
  scale_x_discrete(
    limits = c(
      "Course",
      "Device",
      "Status"
    )
  ) +
  labs(
```

```

    title = "Parallel Sets: Course to Device to Status",
    x = "Stage",
    y = "Students"
  ) +
  theme_minimal()

# -----
# 3. Three largest flows
# -----

top3 <- data %>%
  arrange(desc(Students)) %>%
  mutate(
    Percentage = Students / 250 * 100
  ) %>%
  head(3)

print(top3)

# -----
# 4. Completion rate for each course
# -----

completion <- data %>%
  group_by(Course) %>%
  summarise(
    Total = sum(Students),
    Completed = sum(
      Students[Status == "Completed"]
    ),
    Dropped = sum(
      Students[Status == "Dropped"]
    )
  ) %>%
  mutate(
    Completion_Rate =
      Completed / Total * 100
  )

print(completion)

```

Q5 Interpretation

The largest individual flow is:

Data Science → Laptop → Completed = 45 students = 18%

The three largest individual flows are:

1. Data Science → Laptop → Completed = 45 = 18%
2. Web Development → Laptop → Completed = 35 = 14%

3. Data Science → Mobile → Dropped = 25 = 10%

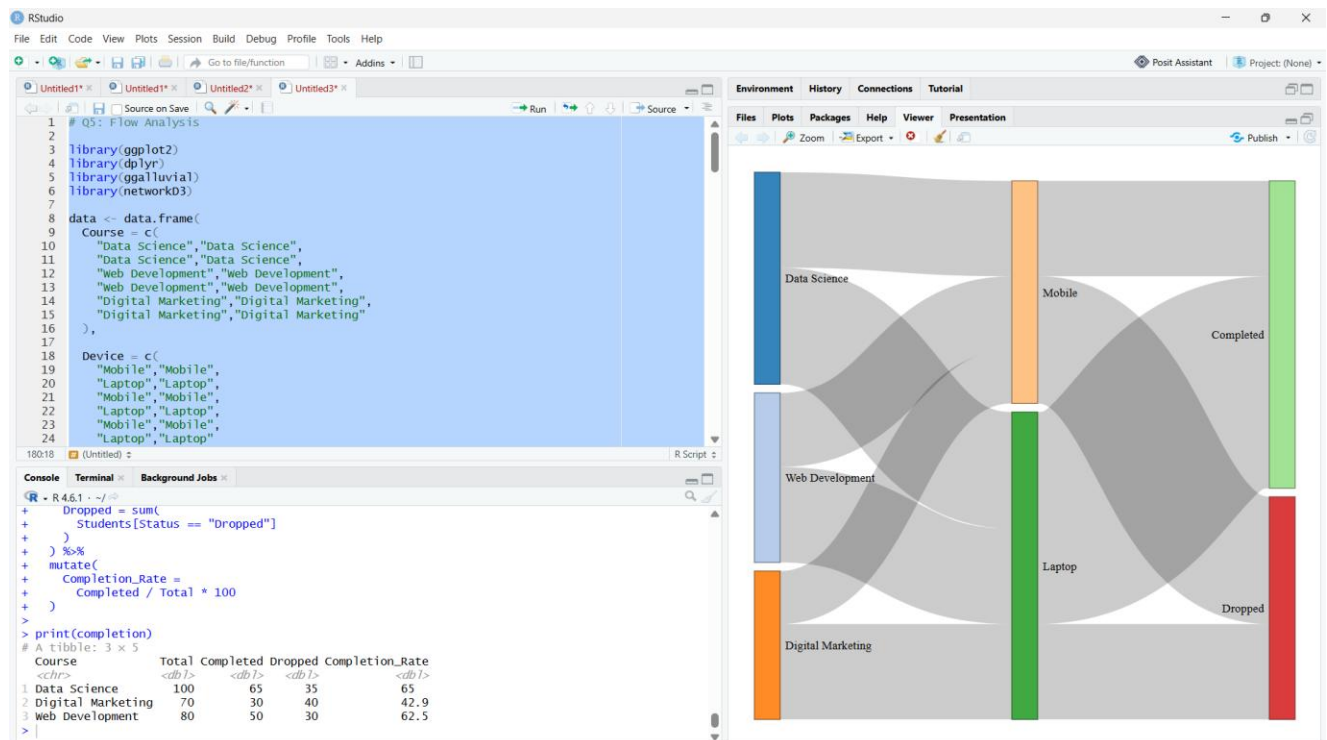
Course completion rates:

- **Data Science:** 65%
- **Web Development:** 62.5%
- **Digital Marketing:** 42.86%

Therefore, **Digital Marketing** has the weakest completion outcome.

Its major problem is particularly visible in the **Laptop → Dropped** pathway, with **25 students dropping**, compared with 20 completing.

A Sankey can become misleading if link widths are not proportional to the actual number of students. Inconsistent node ordering can also make pathways difficult to follow and lead to incorrect comparisons.



Suggested Evaluation Rubric — Data Analysis Project

Criterion	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Data preparation	Correct and complete	Minor issue	Several issues	Incorrect/incomplete
Visualization choice	Highly appropriate and justified	Appropriate	Partially appropriate	Poor choice
Technical execution	Accurate and clear	Mostly accurate	Some errors	Major errors
Interpretation	Insightful and evidence based	Correct	Basic	Incorrect/unsupported
Misleading-design analysis	Bias clearly identified and explained	Bias identified	Limited explanation	Not identified

Suggested marks: 20 marks per question; 100 marks total. Level: Competitive / advanced undergraduate.